

The role of lncRNAs in signaling pathway implicated in CC

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Abstract

Cervical cancer (CC) is the second most common malignancy in females. Owing to poor diagnosis, resistance to the systemic therapies, and high recurrence rate, patients with CC have a relatively poor prognosis. The role of a signaling pathway in CC has always been the focus among worldwide researchers. As reported before, aberrant expression of proteins associated with signaling pathways, such as phosphatidylinositol 3-kinase (PI3K), EGF-R, β -catenin, and Erk and Bcl-2 was discovered in CC. Therefore, aberrant molecular signaling pathways are significant parts of cervical carcinogenesis. Recently discovered long noncoding RNAs (lncRNAs) as a new regulator player of molecular biology in CC have always been reported. In this review, we highlighted the role of lncRNA in signaling pathway implicated in CC and outlined the molecular mechanism of lncRNA in it. All of these present an opportunity for developing diagnosis and therapies against CC.

KEYWORDS

CC, EGFR/VEGFR, ERK/MAPK, lncRNA, PI3K/Akt, signaling pathway, wnt

1 | INTRODUCTION

Cervical cancer (CC), as yet, is one of the major leading causes of cancer deaths in females.¹ Owing to its high recurrence and poor prognosis, there is a great need and urgency in the comprehensive and intensive understanding of the mechanism in CC. It is mainly caused by infection with high-risk human papilloma virus, then involved in the aberrant and uncontrolled cell growth, differentiation, and death due to the integration of viral gene into chromosomal DNA.² High-risk E6 proteins induce the degradation of important cell-cycle regulator,

such as p53 and PDZ domain-containing proteins,³ whereas E7 promotes the inactivation of pRb.⁴ In addition, aberrant expression of proteins associated with signaling pathways, such as phosphatidylinositol 3-kinase (PI3K), EGF-R, β -catenin, Erk, and Bcl-2, was discovered in CC.⁵⁻⁸ Therefore, aberrant molecular signaling pathways are significant parts of cervical carcinogenesis.

Long noncoding RNAs (lncRNAs) are heterogeneous transcripts with more than 200 nucleotides (nt) and no protein-coding function. With the utilization of high-throughput sequencing and other advancing research technologies, the biofunctions of lncRNAs are getting to

Abbreviations: CC, cervical cancer; EGFR, epidermal growth factor receptor; ERK, extracellular signal-regulated protein; HR-HPV, high-risk human papilloma virus; LSD1, lysine-specific demethylase1; MAPK, mitogen-activated protein kinase; PI3K, phosphatidylinositol 3-kinase; PIP3, phosphatidylinositol-3-trisphosphate; PRC2, polycomb repressive complex2; VEGF, vascular endothelial growth factor.

be comprehended. Increasing studies demonstrated that lncRNAs act as important regulators of diverse biological processes.^{9–11} Similarly, increasing evidence has also proved lncRNAs as a new regulator player of molecular biology in CC.^{12,13} Much of the pioneering works in the role of lncRNAs in the signaling pathway implicated in CC still have not been expatiated systematically. In this review, we summarized the origin and function of lncRNA, highlighted the role of lncRNA in signaling pathway implicated in CC, and outlined the molecular mechanism of lncRNA in it. All of these present an opportunity for developing diagnosis and therapies against CC.

2 | ORIGIN AND FUNCTION OF lncRNA

Genome-wide association study confirmed the existence of lncRNA and its biological functions. Most recent release ENCODE 23 indicates that there are 15931 lncRNA present in the human genome.¹⁴ A small part of lncRNAs is recently revealed to exert biological action in proliferation, cell cycle, apoptosis, differentiation, and maintenance of pluripotency at DNA, messenger RNA (mRNA), and protein levels.^{15–17} lncRNA are commonly transcribed from RNA polymerase II and generated from alternative splicing. Afterward, many lncRNAs are reported to encode short peptides. Moreover, much of the pioneering work demonstrated that not only lncRNAs are commonly lowly expressed but also its expression is of temporal and spatial specificity.¹⁸ On the basis of their genomic proximity between neighboring transcripts, lncRNAs are subdivided into five categories: (1) sense lncRNAs, (2) antisense lncRNAs, (3) bidirectional lncRNAs, (4) intronic lncRNAs, and (5) intergenic lncRNAs (also known as lincRNAs).

Although the function of most lncRNAs remains unknown, dozens of research works reported its roles of chromatin remodeling, transcription, and regulation of gene expression in the diverse biological process. The research on epigenetic regulation of lncRNAs started in the early 1990s.^{19,20} Their function has been greatly tied to their modular structure, which permits them to interact with DNA, RNA, and protein. The conformational switching of lncRNAs, whereby their activity is activated or suppressed, renders them as exquisite regulatory devices.²¹ Therefore, their high folding energy distinguishes them from mRNA. lncRNAs are involved in all aspects of gene regulation, including chromosome dosage-compensation, imprinting, epigenetic regulation, nuclear and cytoplasmic trafficking, transcription, mRNA splicing, and translation. It has been accepted broadly that the molecular functions of lncRNA are classified into four archetypes: signaling, guide, scaffold, and decoy.

Signal lncRNAs act as molecular signals to show cellular context or a response to specific stimuli.^{15,21} Guide lncRNAs serve as molecular chaperons, binding specific proteins and then directing the location of the resultant complex to targets. HOTTIP guides the MLL histone H3 lysine 4 methyltransferase complex to the HOXA cluster.²² Scaffold lncRNAs have multiple domains so that they can mediate assembly of multiple molecular components. HOTAIR interacts with polycomb repressive complex2 (PRC2) and lysine-specific demethylase1 (LSD1) complex and the resultant HOTAIR/PRC2/LSD1 complex can suppress the gene expression.²³ Decoy lncRNAs bind and then sequester protein or RNA, in effect sponging up transcription factors and modifiers to regulate the expression of target genes. NEAT2 binds to several splicing factors to alter the pattern of alternative splicing for a set of pre-mRNAs.²⁴

3 | lncRNA IN SIGNALING PATHWAY IMPLICATED IN CC

3.1 | PI3K/Akt signaling pathway

PI3K family participates in a number of signal pathways to regulate many functions of the cell. However, the PI3K/Akt signal pathway is highly anticipated in cancer. There are studies that demonstrated that this pathway is commonly disordered in human cancer and deficiency and mutation of the components of pathway results in proliferation, migration, and transformation of cancer cells.^{25,26} PI3K, composed of an adaptor subunit (p85) and a catalytic subunit (p110), is activated by a combination of Receptor Protein Tyrosine Kinase and receptor, leading to phosphatidylinositol-3-trisphosphate (PIP3) production in the inner membrane. Next, PIP3 recruits a subset of protein with pleckstrin homology domains to the membrane, including PDK1 and Akt. Then, Akt is activated by phosphorylated via PDK1. Once activated, Akt mediates downstream signaling proteins resulting in cellular survival, growth, and migration. There is a major negative regulator, phosphatase and tensin homolog deleted on chromosome ten (PTEN), in the process of the pathway. PTEN is a PIP3 phosphatase, which can dephosphorylate PIP3 to inhibit the PI3K/Akt pathway and make PIP3 inactive.²⁷

The PI3K/Akt pathway is often activated in various human cancer. Recently, the research advances in molecular profiling and genomics have revealed that PI3K/Akt pathway plays a crucial role in cervical carcinogenesis by mediating multiple cellular functions including cell growth, proliferation, metabolism, survival, and angiogenesis.²⁸ Inhibition of PI3K results in a G1 phase cell cycle arrest and there is a synergistic effect of MG132 and LY294002, which act as an inhibitor of the PI3K/Akt pathway on

cervix cancer cell proliferation, apoptosis, and poly (ADP-ribose) polymerase cleavage.²⁹ Schwarz et al⁵ reported alterations in expression of genes from the PI3K/Akt pathway in CC by immunohistochemistry. Furthermore, in the last decade, there has been great progress in studying therapy of CC targeting PI3K/Akt pathway.^{30,31}

Recent evidence suggests that many lncRNAs are involved in PI3K/Akt signaling pathway (Figure 1), thus, the relationship of lncRNA and PI3K/Akt pathway is highlighted. Yang et al³² demonstrated that pAkt is downregulated in the lncRNA UCA1 stable knockdown BLZ-211 cells. Likewise, MALAT1 is verified to activate the PI3K/Akt pathway by increasing the phosphorylation of Akt.³³ However, Xu et al³⁴ suggested that knockdown of MALAT1 in cancer cells increases the levels of phosphorylated Akt. This means MALAT1 is a dual regulator in the PI3K/Akt pathway. Moreover, lncRNA HULC silencing decreases the pAkt by increasing the phosphorylation of PTEN.³⁵ Similarly, Zhu et al³⁶ verified that overexpression of lncRNA LOC572558 is associated with dephosphorylation of Akt, MDM2, and phosphorylation of the p53 protein. Results demonstrated that downregulation of lncRNA H19 significantly enhanced the activity of Akt.³⁷ Furthermore, a study indicated that lncRNA ASLNC20819/22381 could inhibit the combination of IGF and IGF1R to affect the activation of PI3K, which has a negative impact on the Akt pathway.³⁸

Vice versa, PI3K/Akt signaling pathway also regulates the expression of lncRNA. Yu et al³⁹ verified that the expression of lncRNA AK056155 is reduced with treatment of PI3K inhibitor or Akt inhibitor. Moreover, a study indicated that the mRNA level of HOTAIR is decreased when silencing PIK3R3. Meanwhile, the protein and mRNA level of PI3KR3 are decreased when silencing HOTAIR.⁴⁰ This meant that there is an interaction between HOTAIR and Akt pathway.

3.2 | Wnt signaling pathway

The canonical wnt pathway is activated by reception of the secreted wnts, which comprise of a large family of proteins that are highly conserved from fruit fly to human on the seven-pass transmembrane receptors encoded by the frizzled genes.⁴¹ GSK3 β is a major negative regulator of the signaling pathway. Without wnts, GSK3 β formed a multi-protein complex with adenomatous polyposis coli and Axin that can phosphorylate β -catenin for keeping cytoplasmic levels of β -catenin low. Phosphorylated β -catenin is targeted for further degradation by proteasome.⁴² When the signaling is initiated, the multi-protein complex is inhibited and GSK3 β is phosphorylated to an inactive form leading to a block in β -catenin phosphorylation. Thus, cytoplasmic β -catenin is transported into the

nucleus where it regulates target genes expression through partnerships with TCF/LEF.⁴¹

The wnt signaling pathway is related to cell growth regulation, motility, and differentiation. In CC, several studies have shown the aberrant expression of wnt signaling components.^{43,44} Ramos-Solano et al⁴³ demonstrated that restoration of wnt7A in CC-derived cell lines decreased cell proliferation, viability, and migration rates. Likewise, epigenetic silencing of SFRP genes leads to oncogenic activation of Wnt signaling and contributes to CC progression through the EMT program.⁴⁵

Recent studies are beginning to unravel the roles of lncRNAs in EMT and cancer metastasis by regulating the Wnt signaling pathway (Figure 2). Luo et al⁴⁶ demonstrated that H19 activates wnt/ β -catenin by increasing the binding of EZH2 and H3K27me3 of antagonist Nkd1. Likewise, HOTAIR directly decreases WIF-1 expression by promoting its histone H3K27 methylation in the promoter region and then activates the wnt/ β -catenin signaling pathway.⁴⁷ Moreover, lncRNA UCA1 and tsRMST can activate the pathway in a wnts-dependent manner.^{48,49} Chen et al⁵⁰ demonstrated that the overexpression of DANCR significantly decreased the expression levels of p-GSK3 β and β -catenin indicating that lncRNA DANCR can inhibit the activation of the wnt/ β -catenin signaling pathway. Similarly, lncRNA LALR1 activates wnt/ β -catenin signaling by suppressing Axin1.⁵¹ Wang et al⁵² suggested that lnc TCF7 recruits the SWI/SNF complex to the promoter of TCF7 to regulate its expression, leading to the activation of Wnt signaling. Furthermore, CCAL promotes CRC progression by suppressing AP-2 α and further upregulating MDR1/P-gp expression, which, in turn, activates wnt/ β -catenin signaling pathway.⁵³ Recent studies showed that H19 and CASC11 can target hnRNP (heterogeneous ribonucleoprotein) to activate wnt/ β -catenin signaling.^{54,55} Maybe, there are also a large number of lncRNAs that regulate wnt/ β -catenin signal pathway by means we do not know so far.⁵⁶⁻⁵⁸

3.3 | ERK/MAPK signaling pathway

The mitogen-activated protein kinase (MAPK) pathway plays a vital role in crucial cellular processes, including proliferation, differentiation, angiogenesis, and survival. The MAPK family includes extracellular signal-regulated protein, c-Jun N-terminal kinase/stress-activated protein kinase, p38MAPK, and ERK5/big MAP kinase1 pathways. The core of pathway is a three-stage cascade of phosphorylation-dephosphorylation reactions – MAPKKK/MAPKK/MAPK. Once activated, it initiates the three-stage cascade, eventually leading to gene transcription. The most understood downstream is the ERK/MAPK pathway, an evolutionarily conserved signaling

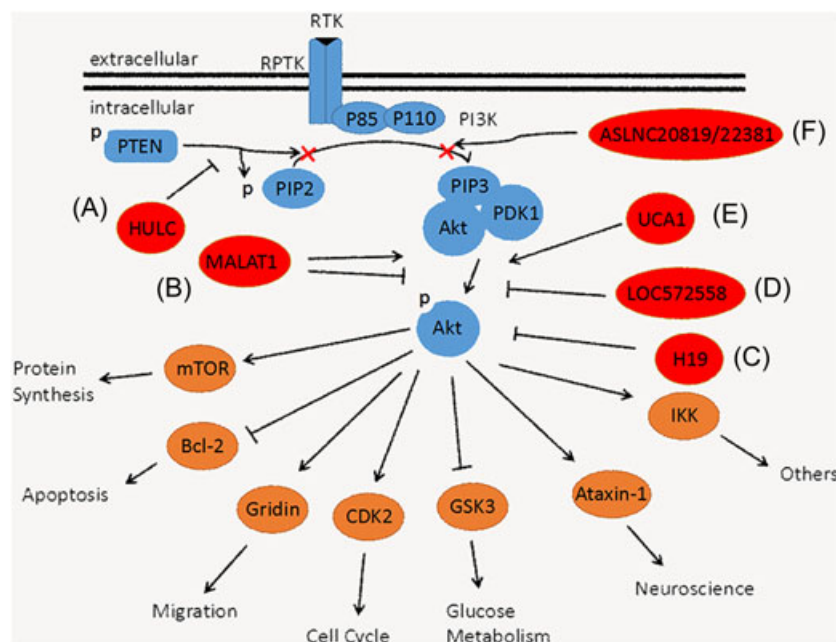


FIGURE 1 Impact of lncRNAs in PI3K/Akt signaling. A, lncRNA HULC silencing decreases the pAkt by increasing the phosphorylation of PTEN. B, MALAT1 is a dual regulator in the PI3K/Akt pathway, which increases or inhibits the phosphorylation of Akt. C, lncRNA H19 significantly degrades the activity of Akt. D, Overexpression of lncRNA LOC572558 is associated with the dephosphorylation of Akt. E, lncRNA UCA1 increases the phosphorylation of Akt. F, lncRNA ASLNC20819/22381 can inhibit the activation of PI3K, which has a negative impact on the Akt pathway

cascade. There are four ways to initiate the cascade, including the binding of receptor tyrosine kinase Ras with ligands, calcium influx through voltage-gated calcium channels or N-methyl-D-aspartate receptors for glutamate, protein kinase C, and G-protein-coupled serpentine receptors.⁵⁹⁻⁶² Initiation of Ras leads to sequential activation of Raf, MEK1/2, and subsequently ERK1/2. In cytoplasm and nucleus, there are plenty of substrates of activated ERKs. Activated ERKs can stay on the cytoplasm to phosphorylate cellular skeletal protein and activate other protein kinases or are transported into the nucleus to phosphorylate gene transcription factor.⁵⁹⁻⁶³

The pioneering endeavor revealed the vital role of ERK/MAPK signaling pathway in carcinogenesis and therapy of CC. Human papilloma virus (HPV) 16 E5 plays an important role in the early stage of CC via activating MEK/ERK1/2.⁶⁴ It also demonstrated that the alteration of this pathway presents an effective role in the chemosensitivity of CC-derived cells.⁶⁵⁻⁶⁷ Recently, the therapeutic function of MEK1/2 in CC has been into the clinical protocol.⁶⁸

In view of the critical part of the ERK/MAPK pathway in the treatment of CC, likewise, inside lncRNAs are indispensable (Figure 3). Knockdown of lncRNA RMEL3 and URHC can regulate cell proliferation and induce apoptosis through inactivation of ERK/MAPK pathway.^{69,70} Higher levels of phospho-ERK1/2(p-ERK1/2) and p-MEK were observed in DBH-AS1 and BANCER overexpressing cells than in control cells,⁷¹⁻⁷³ and, CCHE1 downregulation significantly reduced the levels of p-MEK1/2, p-ERK1/2 and p-MAPK.⁷⁴⁻⁷⁶ Furthermore, Wang et al⁷⁷ reported that the knockdown of lncRNA BANCER greatly inhibited the ERK/

MAPK pathway by decreasing MMP1 and MMP2. Simultaneously, a study by Han et al⁷⁸ demonstrated that MALAT1-mediated tumor suppression may be due to the reduction of ERK/MAPK signaling pathway activity by downregulation of MMP2. Interestingly, it also showed the contrary results that upregulation of MALAT1 significantly reduced the levels of p-ERK1/2. This contradiction meant that MALAT1 played different roles in diverse cancers. In addition, the expressions of lncRNA and molecules of the ERK/MAPK pathway interacted with each other. The results indicated that the mRNA and protein level of MAPK1 was decreased when silencing HOTAIR and the mRNA level of HOTAIR was decreased when silencing MAPK1.⁷⁹

3.4 | EGFR/VEGFR signaling pathway

Epidermal growth factor receptor (EGFR) is involved in the ErbB signaling pathway. EGFR interacts with itself or ErbB family to form a homodimer, heterodimer, or oligomers. The ErbB family of membrane-bound tyrosine kinase receptors consists of Egfr, ErbB2, ErbB3, and ErbB4, which activate potent signaling pathway. Upon ligand binding to the extracellular domain of EGFR, autophosphorylation of EGFR to phosphorylated EGFR (pEGFR) leads to activation of the downstream pathway: Ras-MAPK, PI3K-AKT, Src, STAT3 and so on, which are important for cell proliferation, differentiation, motility, and survival.⁸⁰⁻⁸² EGF and EGFR have been studied previously in CC. EGF induces cell migration in human CC-derived SiHa cells by the

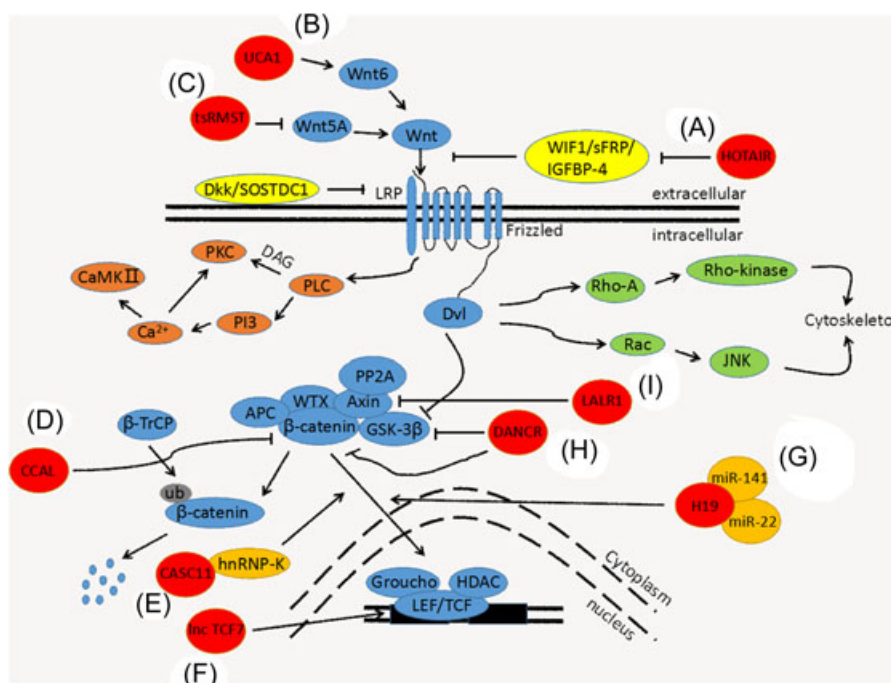


FIGURE 2 Impact of lncRNAs in Wnt signaling. A, HOTAIR directly decreases WIF-1 expression by promoting its histone H3K27 methylation in the promoter region and then activates the wnt/ β -catenin signaling pathway. B, UCA1 can activate the pathway by increasing Wnt6. C, lncRNA tsRMST can activate the pathway by degrading Wnt5A. D, CCAL promotes CRC progression by suppressing AP-2 α and further upregulating MDR1/P-gp expression, which, in turn, activate wnt/ β -catenin signal pathway. E, CASC11 can target hnRNP to activate wnt/ β -catenin signaling. F, lnc TCF7 recruits the SWI/SNF complex to the promoter of TCF7 to regulate its expression, leading to activation of Wnt signaling. G, H19 directly activates the pathway by combination with microRNA. H, DANCR significantly decreases the expression levels of p-GSK3 β and β -catenin and then inhibits the activation of the wnt/ β -catenin signal pathway. I, LALR1 activates wnt/ β -catenin signaling by suppressing Axin1

EGFR/Erk 1/2 signaling transduction pathway.⁸³ The HPV oncoproteins have been demonstrated to be linked to the activation of EGFR.^{84,85} However, the prognostic significance of EGFR pathway in CC is equivocal. Immunohistochemical and multivariate analysis showed that EGFR gene amplification was an independent prognostic factor for overall survival.⁸⁶ EGFR and pEGFR are independently associated with poor response to therapy and disease-specific survival in patients primarily treated by (Chemo)radiation.⁸⁷ Interestingly, Eijsink et al⁸⁸ reported the expression of EGFR pathway components, which does not appear to have prognostic impact in surgically treated early-stage CC.

The vascular endothelial growth factor (VEGF) family is a class of cytokine having a potent role in angiogenesis. The VEGF family includes VEGF-A~D and placenta growth factor in mammals and two exogenous VEGF subtypes, virus genome-encoded VEGF and snake venom-derived VEGF. Angiogenesis is triggered through VEGF binding to members of the VEGFR family, including VEGFR1 (Flt-1), VEGFR2 (KDR/Flk-1), and VEGFR3 (Flt-4). Among these receptors, VEGF binding to VEGFR2 is considered as the most important pathway for angiogenesis, and VEGFR2 can

bind all VEGF-A isoforms, VEGF-C, and VEGF-D. The downstream pathway includes the Raf-MAPK pathway, the PI3K-AKT pathway, and so on. Recently, VEGF/VEGFR are reported to induce the formation of the capillary vessel in tumor cells, cellular proliferation, and endothelial cell survival.⁸⁹⁻⁹¹ In CC studies, adjacent CIN3 displayed strong expression of VEGF-C, VEGF-D, and VEGFR3.⁹² VEGF-C promoted CC metastasis and invasion by Flt-4/c-Src and VEGFR/NF- κ B pathway.^{93,94} Multivariate analysis showed that the expression of VEGFR2 is related to the prediction of radio-chemotherapy in CC⁹⁵ and higher VEGFR1/2 expression levels were predictive of poor overall survival⁹⁶ and EGFR may mediate HPV16 E6-induced initiation and progression of CC.⁹⁷

However, there are only few studies exploring the relationship of lncRNA and EGFR/VEGFR pathway (Figure 4). Su et al⁹⁸ reported that the levels of MEG3 are inversely correlated with VEGF levels. Conversely, XIST is positively correlated with EGFR.⁹⁹ In addition, overexpression of GAS5 was inversely correlated with the expression of the EGFR pathway.¹⁰⁰ Furthermore, Zhang et al¹⁰¹ found that MET/EGFR pathway was regulated by lncRNA FAM83H-AS1. Thus, speculating FAM83H-AS1

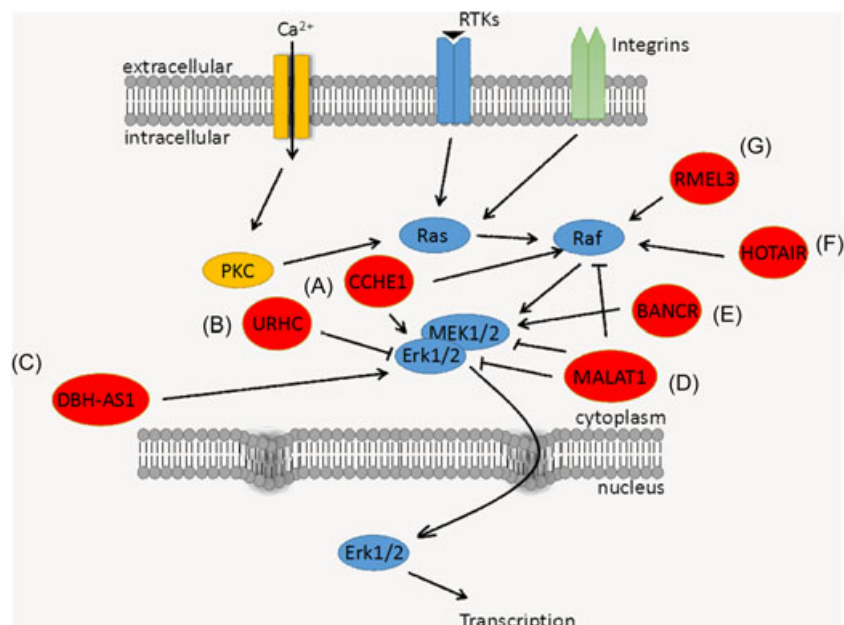


FIGURE 3 Impact of lncRNAs in ERK/MAPK signaling. A, CCHE1 can significantly increase the levels of p-MEK1/2, p-ERK1/2, and p-MAPK. B, URHC can inhibit the ERK/MAPK pathway by degrading ERK1/2. C, DBH-AS1 can increase the expression of ERK1/2 to activate the ERK/MAPK pathway. D, MALAT1 significantly reduced the levels of p-MEK1/2, p-ERK1/2, and p-MAPK. E, BANC1 can increase the expression of MEK1/2 to activate the ERK/MAPK pathway. F, HOTAIR can increase the level of MAPK to activate the ERK/MAPK pathway. G, RMEL3 can increase the level of MAPK to activate ERK/MAPK pathway

may provide a novel therapeutic target impacting MET/EGFR pathway. In view of the internal mechanism, lncRNA CAR10 upregulates EGFR by binding to and stabilizing transcription factor YB-1.¹⁰² ANRIL mediates the PRC2 components p300 and miR200b and directly

binds to p300 and EZH2, leading to the regulation of VEGF expression and function.¹⁰³ And, the EGFR gene CpG island methylation significantly increased by depleting lncRNA AK123072, resulting in the down-regulation of EGFR.¹⁰⁴ Similarly, EGFR/VEGF pathway

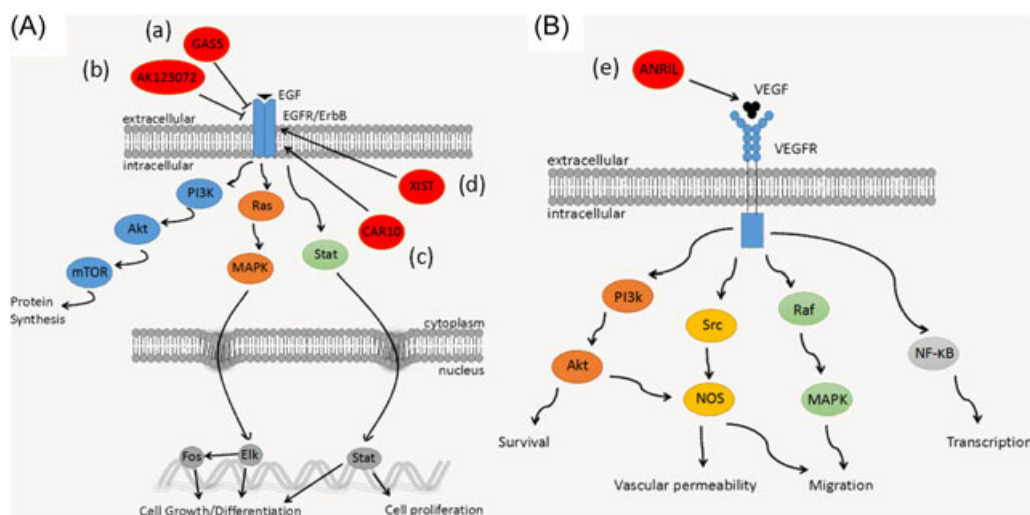


FIGURE 4 Impact of lncRNAs in EGFR/VEGFR signaling. A, GAS5 degrades the expression of the EGFR. B, The EGFR gene CpG island methylation significantly increased by depleting lncRNA AK123072 resulting in the downregulation of EGFR. C, CAR10 upregulates EGFR by binding and stabilizing transcription factor YB-1. D, XIST increases the expression of the EGFR. E, lncRNA UCA1 increases the phosphorylation of Akt. F, ANRIL mediates the PRC2 component p300 and miR200b and directly bind to p300 and EZH2, leading to the regulation of VEGF expression and function

can regulate the expression of lncRNA. EGF down-regulates lncRNA LIMT expression by enhancing histone deacetylation at the promoter.¹⁰⁵

4 | CONCLUSION AND PERSPECTIVE

Despite papanicolaou smear, colposcopy programs, surgery, radiation and chemotherapy, CC still contributed as the second highest number of deaths in female cancers carrying high risks of morbidity and recurrence. Thus, it is important to develop early diagnostic approaches and more effective treatments. Molecular pathway that participates in cervical malignant transformation has emerged as promising therapeutic targets. lncRNA plays an important role in signaling transduction in cervical carcinogenesis and could be used as potential biomarkers or targets for CC diagnosis, treatment, prognosis, and relapse via signaling pathway in mechanisms of interaction with protein/mRNAs. However, the exact molecular mechanisms of lncRNAs in CC remain unclear. Studies elucidating these complicated mechanisms and clinical applications are needed to be explored and validated further.

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CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

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